VSM Base: User's Guide

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1 Introduction

VSM Base is a parametric model of a portable, hinged base for vintage flatbed sewing machines. You give it measurements of your machine's bed casting; it produces a base whose seat, hinge recesses and clearances fit that casting, and that prints in two halves on an ordinary desktop printer.

1.1 Why this exists

Many vintage sewing machines were sold built into cabinets and tables. Today few people have room for a sewing table, and a machine without a portable base is hard to use, to store, or even to give away. Too many good machines are scrapped for that reason alone.

Collectors make their own bases, and a few small makers sell them. The difficult part is the hinge recesses and the fit of the bed: they need careful work with a router and chisels, and so most homemade bases are boxes the machine simply sits in. The shops and skills for that kind of work are becoming scarce. 3D printers, on the other hand, are now common and inexpensive, and a printer is very good at exactly the precise, repeatable shapes that are hard to make by hand.

The model is published under the MIT license so that nobody has to repeat the work. It is only useful, though, if it works the first time: a full base is a long print and a lot of filament. This guide is written to make that likely. It asks you to measure carefully, to read what the model tells you about your numbers, and to print a few small test parts before the real thing.

1.2 How to use this guide

Work through the chapters in order the first time:

- **What you need**: printer, filament, software and hardware.
- **Measuring your machine**: the measurements the model needs, how to take them, and a worked example for the Singer 99K.
- **Configuring the model**: the Customizer, which settings matter for fit, and how to read the console output.
- **Test parts**: what each one checks, what a good result looks like, and what to adjust if it isn't.
- **Printing**: orientation, supports, and the features that are removed or drilled out after printing.
- **Assembly**: plates, nuts, hinges, rods, extensions and lids.
- **Other machines and hinges**: adapting the model and contributing a profile.

The last chapter is a reference listing every parameter in the model. It is generated from the model's own comments, so it always matches the code.

1.3 Conventions

Directions. Directions are given as you sit at the machine to sew. *Front* is the side nearest you and *back* is the hinge side. On most flatbed machines, including the Singer 99, the needle is at the *left* end and the pillar and handwheel are at the *right*.

Units. Measurements of the machine's bed are entered in **inches**, because vintage castings were designed in fractional inches and are easiest to measure that way; enter them as decimals (12 3/16" is `12.1875`). Everything else in the model, including the casting reliefs and bobbin trenches, is in **millimetres**. Each parameter's description says which.

Parameter names are shown like `bed_length_in`. They are the names you will see in OpenSCAD's Customizer and in the parameter reference.

Coordinates. Inside the model, X runs from left to right, Y from front to back and Z upward, with the origin at the base's outer front-left bottom corner. You only need this if you read the code.

2 What you need

This chapter is an outline, still to be written. It will cover:

- Printer: build volume needed for each part, and how to check it with `printer_size`.
- Filament: PETG (tested), why not PLA for a base that may sit in a warm room or a car, and how much to allow.
- Software: OpenSCAD 2021.01 or later, installing BOSL2, and a slicer.
- Hardware list with quantities and sources: hinges, mending plates, screws, nuts, rods, cushions.
- Tools: calipers, drill bits for the membranes, pliers for the knock-out walls.

3 Measuring your machine

This is the most important chapter in the guide. Every critical dimension of the base is computed from a handful of measurements of your machine's bed casting. If they are right, the base fits; if one is wrong, the base will be wrong in a way no amount of printer tuning can fix.

3.1 Measure the casting, not an old base

If your machine came in an old wooden base, it is tempting to copy that. Don't. Wooden bases were hand-fitted, they shrink and wear over decades, and many have been swapped between machines: a base that came with your 99 may have been made for a 128, or fitted by eye in a shop sixty years ago. The cast-iron bed of the machine has not changed since it left the factory. Measure that.

3.2 Tools

- **Calipers** (dial or digital), 6" or larger. These are essential for everything under an inch: plate thickness, hole position, hole diameter and the thickness of the casting's bosses.
- **A steel rule**, 12" or longer, graduated in 32nds or 64ths, for the bed's length and width.
- **A small square** is useful for measuring the hinge-hole height.
- Paper and pencil. Write every number down as you measure it; the worksheet at the end of this chapter has a place for each one.

Measure everything at least twice. Where the guide asks for the same measurement at both ends of the bed, take both and compare them.

3.3 Getting ready

Remove the machine from any base or cabinet, take off the motor belt guard or anything else that hangs below the bed if it's easy to remove, and work on a clean, flat table. Several measurements are easiest with the machine on its back (hinge side down) or upside down on a towel.

Fractions to decimals, for entering the bed measurements:

16ths	inch	16ths	inch	16ths	inch	16ths	inch
1/16	0.0625	5/16	0.3125	9/16	0.5625	13/16	0.8125
1/8	0.125	3/8	0.375	5/8	0.625	7/8	0.875
3/16	0.1875	7/16	0.4375	11/16	0.6875	15/16	0.9375
1/4	0.25	1/2	0.5	3/4	0.75	1	1.0

3.4 The bed outline



Figure 1: Top view of the bed. The bed is the flat casting the machine's arm stands on; its outline sets the size of the seat in the base.

3.4.1 Bed length, `bed_length_in`

Measure the bed from end to end, left to right, along its top surface. Measure along a straight part of the edge, not across the rounded corners, and read the rule at the very ends of the casting, not at any paint lip or decal.

The base's seat is made `bed_clearance` (0.75 mm) larger than the bed on each side, so the bed drops in without binding. Don't add your own allowance: enter what you measure.

3.4.2 Bed width, `bed_width_in`

Measure the bed from front to back in the same way, near the middle of the bed where the front and back edges are straight.

3.4.3 Plate thickness, `bed_plate_thickness_in`

The bed is a flat plate with a skirt and bosses cast underneath. Measure the thickness of the flat plate with calipers at a front edge or an end, away from any thickened area. On the Singer 99 it is 5/16".

The plate thickness, together with the hinge-hole height below, determines how deep the hinge pins must sit in the base, so it has to be right.

3.5 The hinge holes

The machine pivots on two pins that enter holes in the rear skirt of the bed. The model places the hinge recesses so that the pins line up with those holes exactly, both along the back and in height.



Figure 2: Rear of a Singer 99K bed, with a rule along the back. The hinge holes are the two holes in the rear skirt, one near each end.

3.5.1 Hole position, `hinge_hole_inset_in`

Measure from each end of the bed to the **center** of the nearer hinge hole. The center can't be measured directly, so measure from the end of the bed to the near edge of the hole and to the far edge, and average the two; or measure to the near edge and add half the hole's diameter.

Do this at both ends. The model assumes the holes are the same distance from each end. On the 99 they are 1 7/16" from each end.

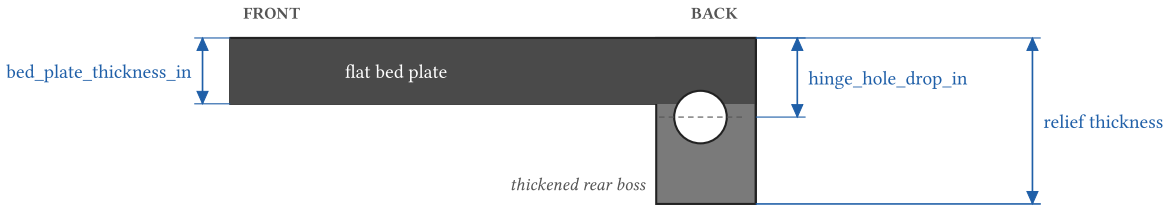
Check your work. Also measure the distance between the two hole centers. The two insets plus the spacing between the holes must add up to the bed length:

> inset + spacing + inset = length
>
> On the 99: 1 7/16" + 9 5/16" + 1 7/16" = 12 3/16". ✓

If they don't add up, measure again before going any further. If the two insets really differ by more than about 1/32", the model's symmetric placement won't fit your machine; see the chapter on other machines.

3.5.2 Hole height, `hinge_hole_drop_in`

Measure how far the **center** of the hinge hole is below the **top surface** of the bed. With the machine on its back, lay a small square or a straightedge across the bed's top surface where it overhangs the hole, and measure from it to the top and bottom edges of the hole; the center is halfway between. On the 99 it is 3/8".



Section through the back edge at a hinge hole, seen from the left end.

Figure 3: Section through the back edge at a hinge hole. On the 99 the hole sits below the flat plate, in a thickened boss.

3.5.3 Hole diameter

Measure the hinge holes' diameter. It isn't a model parameter, but it is the diameter of the hinge pins you need. On the 99 the holes are 1/4".

3.5.4 Bed cushions, `bed_cushion_in`

Most bases have small rubber or felt cushions under the corners of the bed, which quiet the machine and keep it from rocking. The model lifts the bed by their thickness when it works out the hinge height. Enter the thickness of your cushions *when compressed* under the machine's weight; the default is 1/16". If you use no cushions, enter 0.

Fit the cushions before you check anything against the machine: without them the bed sits lower than the model expects, and the extensions will stand proud of it. Cushions also vary in how much they compress. Once the base is printed and the machine sits in it on its cushions, lay a straightedge from the bed onto an extension (or its top slice from the test parts). If the extension is high or low, set `ext_top_adjust` to the difference (negative lowers it); it moves the whole extension top plane, lids, lips and bobbin trenches together.

Once these numbers are entered, the console reports the height of the bed's hole axis and the height of the hinge pins' axis. They must agree: if they differ by more than a quarter of a millimetre the model prints a WARNING. Fix that before printing anything.

3.6 Thickened areas of the casting, `bed_reliefs`

Around its edge, the bed rests on a ledge in the base. Most castings are thickened in places along the edge: bosses around the hinge holes, ribs at the ends, lugs for screws. Wherever the casting hangs below the flat plate over the ledge, the ledge must be cut away under it, or the bed won't sit down. These cut-aways are the **reliefs**.

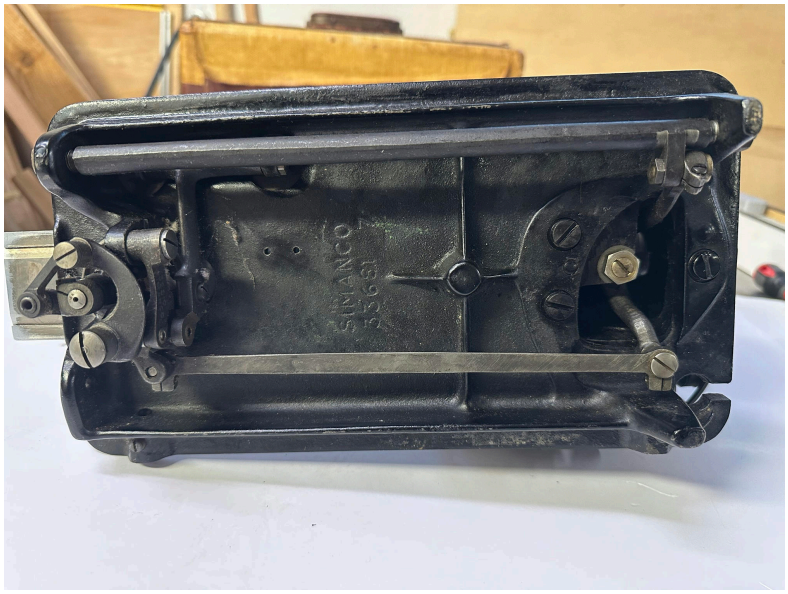


Figure 4: Underside of a Singer 99K. The flat plate, the rear skirt with the hinge bosses, and the thickened centers of each end are visible around the edge.



Figure 5: How the thickened areas are measured. Each one becomes an entry in `bed_reliefs`.

Turn the machine upside down and look along each edge. For each place where the casting is deeper than the flat plate within about 10 mm (3/8”) of the edge, measure:

- **Which edge** it is on: `"left"`, `"right"`, `"back"` or `"front"`.

- **Where it starts and where it ends** along that edge. Measure along the back and front edges from the bed's *left end*, and along the left and right ends from the bed's *front edge*. The **center** is halfway between the start and the end, and the **length** is the end minus the start.
- **How thick** the casting is there, measured from the bed's top surface down to the lowest point of the boss. At the edge of the bed you can usually reach both surfaces with the outside jaws of the calipers.



Figure 6: The left end of a Singer 99K from below, showing the thickened center of the end.

`bed_reliefs` is in **millimetres** (inches × 25.4). Each entry is `[edge, center, length, thickness]`, and the whole setting is a list of entries:

```
bed_reliefs = [["left",  83.34, 76.2, 15], ["right", 83.34, 76.2, 15],
               ["back",  36.51, 50.8, 20], ["back", 273.05, 50.8, 20]];
```

The model makes each relief a little bigger than you measured: 2 mm longer at each end (`relief_margin`) and 1 mm deeper (`relief_clearance`). You don't need to add your own allowance, but when an edge is hard to see, it is better to measure generously than tightly. A relief that is too big is invisible once the machine is in the base; one that is too small stops the bed from seating.

The console lists each relief with its position and depth, so you can check that it landed where you meant.

3.7 Room under the bed

The base's cavity has to clear everything that hangs below the bed: the hook or shuttle drive, the feed mechanism, the shafts and their bearings. With the defaults, the cavity is about 295 × 152 mm in plan, set 8 mm in from the bed's edge all round (`bed_ledge`), and there are about 75 mm (just under 3") from the bed's top surface to the cavity floor.

With the machine upside down, lay a straightedge across the bottom of the bed and measure how far the lowest part of the mechanism stands above it; add the plate thickness to get its depth below the bed's top surface. Check, too, that nothing hangs below the bed within 8 mm of its edge except the bosses you entered as reliefs. If anything is too deep, increase `inner_depth`; if anything is too close to the edge, reduce `bed_ledge` or add a relief.

The center slice in the test parts (see the chapter on test parts) has the full inside profile of the base, and is the best check: slide it along the underside of the machine and see that nothing touches.

3.8 The bobbin cover, `left_trench` and `back_trench`

On drop-in bobbin machines the bobbin is reached through a slide plate in the bed, which slides out past the edge of the casting. The base itself doesn't get in its way, because the bed stands above the base's top surface. The extensions, though, are level with the bed, so an extension on that side needs a shallow trench for the plate to slide into.

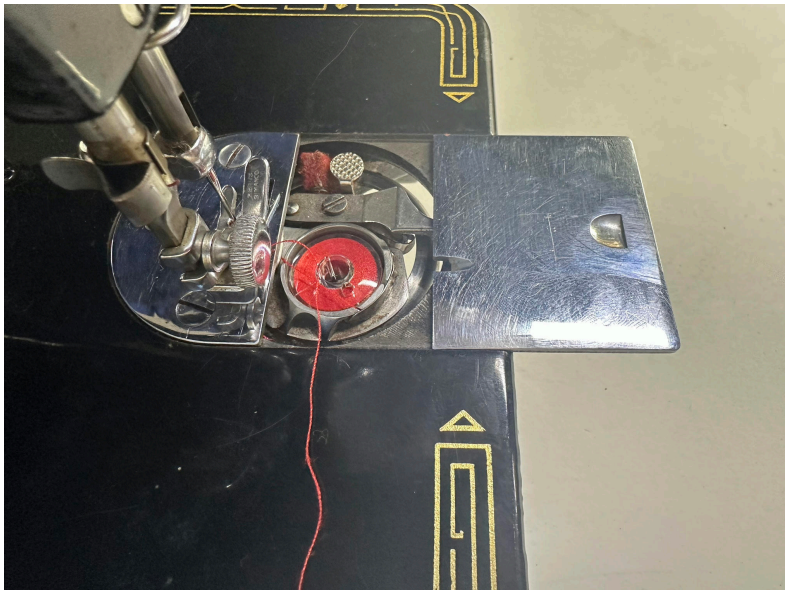


Figure 7: A Singer 99K's slide plate pulled open. It slides out past the left end of the bed.

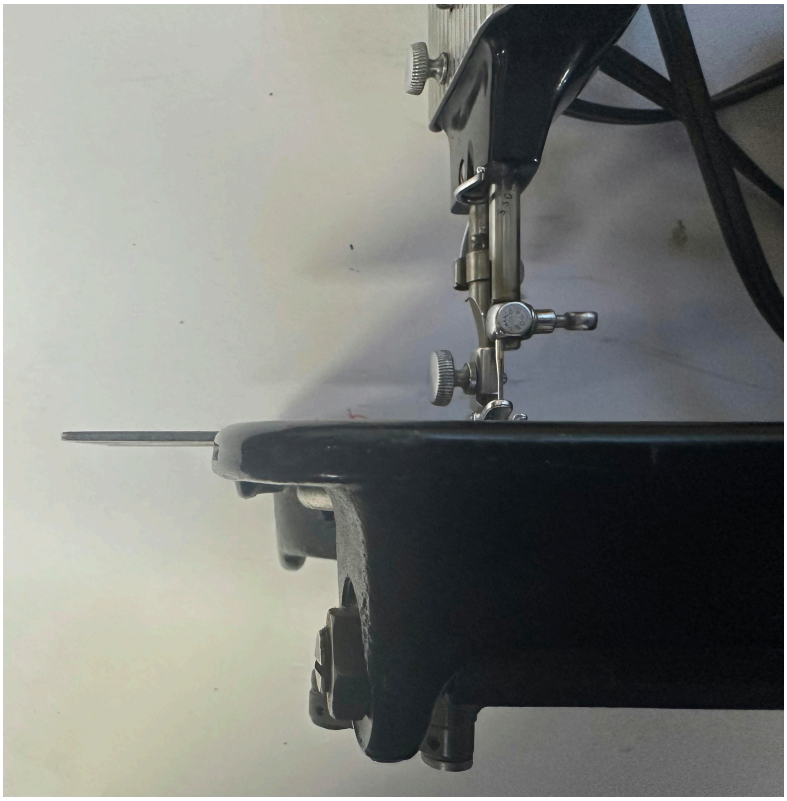


Figure 8: Seen edge-on, the plate stays level with the bed's top surface as it slides past the edge of the casting.

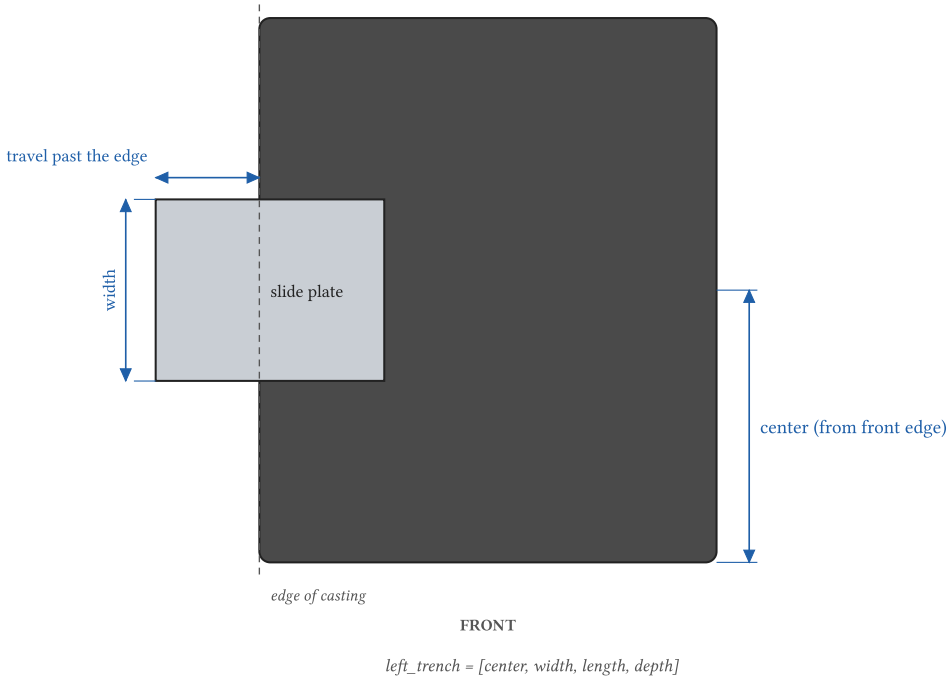


Figure 9: How the bobbin-plate trench is measured.

Measure, in millimetres:

- **Center**: the middle of the plate across its width, from the bed's front edge (for `left_trench`) or from the bed's left end (for `back_trench`).
- **Width**: the plate's width plus about 1 mm of clearance each side.
- **Length**: how far past the edge of the casting the plate must slide to uncover the bobbin, plus a margin. Allow at least 1/4" more than you measure.
- **Depth**: the plate's thickness plus about 0.5 mm.

On the Singer 99 the plate is 2 3/16" wide, centered on the left end, 2 mm thick, and slides about 1 1/4" past the casting, giving `left_trench = [83.34, 57.6, 40, 2.5]`. On the Singer 128 the plate slides toward the back, and `back_trench` is used instead. Set a trench to `[]` if your machine doesn't need one.

3.9 Worked example: Singer 99K

These are the defaults in the model, measured from a 1950s Singer 99K. Your 99 should be very close, but measure it anyway.

Measurement	Measured	Parameter	Entered
Bed length	12 3/16"	`bed_length_in`	`12.1875`
Bed width	6 9/16"	`bed_width_in`	`6.5625`
Plate thickness	5/16"	`bed_plate_thickness_in`	`0.3125`
Hinge hole from each end	1 7/16"	`hinge_hole_inset_in`	`1.4375`
Hinge hole below bed top	3/8"	`hinge_hole_drop_in`	`0.375`
Bed cushions, compressed	1/16"	`bed_cushion_in`	`0.0625`
Hinge hole diameter	1/4"	(hinge pins)	—
End bosses, middle 3" of each end	15 mm thick	`bed_reliefs`	`["left", 83.34, 76.2, 15], ["right", 83.34, 76.2, 15]`
Hinge bosses, 1" either side of each hole	20 mm thick	`bed_reliefs`	`["back", 36.51, 50.8, 20], ["back", 273.05, 50.8, 20]`
Slide plate	2 3/16" wide, 2 mm thick, 1 1/4" travel	`left_trench`	`[83.34, 57.6, 40, 2.5]`

Every measurement on this machine is an odd number of sixteenths of an inch, and the hinge-hole check adds up exactly: a good sign that the numbers are right.

3.10 Checking your numbers

After entering your measurements, render the base (F5) and read the console. In particular:

- **`Bed seat`**: should be your bed's length and width plus 1.5 mm.
- **`Bed hole axis Z` and `pin axis Z`**: must be equal. A WARNING here means the hinge pins won't line up with the bed's holes.

- **`Machine cavity`**: the clear space under the bed, in plan.
- **`Relief`** lines: one for each thickened area, with its position and depth.
- **`Base part`**: the size of each printed half, with a WARNING if it won't fit your printer (`printer_size`).

When everything looks right, go on to the test parts. The first one to print is the top slice of the base: drop the machine into it and check the fit before printing anything else.

3.11 Measurement worksheet

Measurement	First	Second	Entered
Bed length (in)			
Bed width (in)			
Plate thickness (in)			
Hinge hole inset, left (in)			
Hinge hole inset, right (in)			
Hinge hole spacing (in)			
Hinge hole below bed top (in)			
Hinge hole diameter (in)			
Bed cushions, compressed (in)			
Thickened area 1: edge, start, end, thickness			
Thickened area 2			
Thickened area 3			
Thickened area 4			
Lowest point under the bed, from bed top (mm)			
Slide plate: center, width, travel, thickness			

4 Configuring the model

This chapter is an outline, still to be written. It will cover:

- Opening the model and using OpenSCAD's Customizer.
- The settings that decide the fit (machine bed, reliefs, hinges) and the ones that are a matter of taste (compartment, extensions, rounding).
- The `part` setting and the views it gives: assembly, all, test parts, individual parts.
- Reading the console: what each line means, and every WARNING with what to do about it.
- Choosing a side compartment, extensions, trenches and lid hinges.
- Organizer inserts: bobbin trays (corner bay), notion trays and the spool tray (back or left bay, chosen with `ot_bay`), the stack-height budget the console reports, and the `*_assembly` views.

Note: an outlet base used for storage first. Set `compartment_option = "outlet"` and `passthrough_knockouts = true`: the base gets the outlet's curbs and screw hole, but the cord pass-through is closed at both ends by thin walls, so the compartment works for storage. Cover the outside screw hole with the printed cap. For a storage lid, switch `compartment_option` to `storage` just to print the lid; it is the same size. When you fit the outlet, tap the two walls out with a punch and a small hammer, and print the outlet lid. To close the pass-through again later, print `passthrough_cover`.

Note: a short back extension. If `back_ext_length` is too short to reach the last back peg hole, the back lip runs on past the body over the base's back rim to reach it (`back_lip_extend`), with a rounded end and a fillet into the body; the console says how far. That extension then fits only a base with the same `back_peg_xs`. If you move the base's peg holes left with `back_peg_xs` instead, the lip needn't run on, but the extension, again, fits only bases with those holes.

Note: outlet labels. With the outlet, LIGHT (always on) and MOTOR (the controlled output) are recessed into the floor by their receptacles, LIGHT at the back by default (`outlet_light_side`), since the outlet goes in with the always-on side and the cord to the back. The words, size and depth are settings; `outlet_labels = false` leaves them off.

Note: insert trays need their bay. The bobbin trays need the corner bay and the notion and spool trays need the bay named by `ot_bay`. The bay can be the attached extension or its freestanding box (`make_corner_box` and so on), which is how to hold a second bobbin stack. With no bay, the render stops with a message saying which setting to turn on.

Note: notion tray heights. The console's STACK line gives the total height the trays may add up to in the chosen bay (78.2 mm in the back bay at the defaults), so any mix of the 30, 25, 20, 15, 10 mm and custom trays that totals no more than that fits under the lid. The spool tray line gives what is left above a spool tray.

5 Test parts

This chapter is an outline, still to be written. It will cover:

- Why print tests, and the order to print them in.
- Top slice of the base: does the bed drop in and sit flat, and do the hinge pins line up?
- Dovetail coupon: joint clearance (`split_clearance`, `key_flare_angle`) and the plate-pocket bridge.
- Center slice: all three dovetails with the plates installed, the front seam, and clearance under the machine.
- Left-end strip: handhold feel, knock-out wall, rod fit (`rod_hole_d`).
- Hinge block: hinge fit and bolt length.
- Hinge coupon (also `part = "hinge_coupon"`): a quick check of the hinge recess and pin slot without printing a slice of the base.
- Center slice: the front-wall serrations, and which half goes on first (lower the right half onto the left).
- Rod plugs: press fit in the end strip's rod holes.
- Split test (its own part, `part = "split_test"`, not in the test set): just the front of the split joint, both halves, with the front-wall serrations and the first dovetail key, for a quick check of a serrated joint. The wall stubs are `split_test_h` tall (30 mm; 0 for full height).
- Outlet block (outlet compartment only): curbs, screw height, and the insert or through-screw hole, with your outlet and plugs in place.
- Extension top slices: lip fit, pegs, supports, and the bobbin-plate trench.
- Lids: snap-in hinge, swing, and the lean-back stop.

> **Note: the hinge coupon as a wooden-base insert.** The hinge coupon has countersunk corner holes for #8 flat-head wood screws, so it can also be used in a base made of wood. Print two, mortise them flush into the top of the base with their front edges on the back wall of the bed opening, at the hinge spacing the console gives, and screw them down. The console also gives how far below the bed's top surface the coupon's top must sit, and where the hinge-pin axis falls. The router work for the hinge recesses, the hardest part of a wooden base, is then done by the printer.

6 Printing

This chapter is an outline, still to be written. It will cover:

- Slicer settings that matter: walls, infill, bridging, and brims.
- The base halves: orientation and what to expect.
- Extensions: printed upright with supports under the lips.
- Print plates: `part = "all"` gives every part of the build in print orientation, grouped into plates for your `printer_size` and `printer_margin`; `print_plate = N` gives plate N on its own. The console lists the plates and which need supports.
- Lids: printed top face down. Print `lid_hinge_test` first to check the snap-in hinge in your filament.
- Extension pegs (`part = "ext_pegs"`): printed lying down, never standing up; print a few spares.
- Features removed after printing: knock-out walls under the handholds, sacrificial membranes over screw holes.

Note: lid hinges in PLA and PETG. The hinge defaults (`lid_tab_w = 4`, `lid_bump_engage = 1.0`) come from test prints in both PLA and PETG. With the narrower 2.5 mm flex tabs of earlier versions, a PLA lid popped out of its hinge almost every time it was opened: PLA is stiffer than PETG and the narrow tab bowed away instead of holding the bump in. If a lid still comes out too easily, tighten `lid_hinge_end_clearance` or make the bumps steeper with a smaller `lid_bump_angle`; if it is hard to snap in, do the opposite. Try changes on `lid_hinge_test`, which prints in a few minutes, before printing a full lid.

Note: supports under the lips and trenches. The extension lips and the bobbin-plate trenches in the lids need supports. Standard (not tree) supports, with a separate support material printed on the interface layer, give a clean underside: the trench floor below was printed over them, in a lid printed top face down so the top takes the bed's texture. On a printer with a second material, the slicer adds a purge tower to each plate with supports; leave room for it.

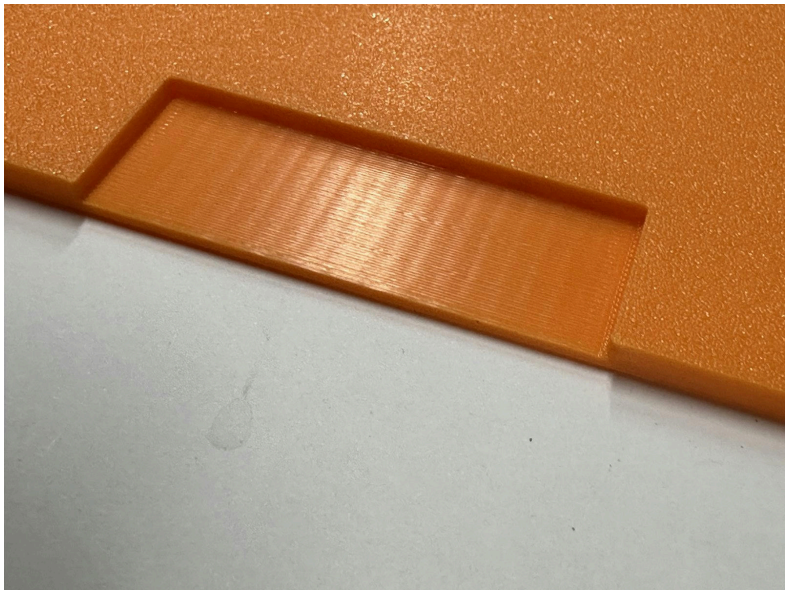


Figure 10: The floor of a bobbin-plate trench, printed over standard supports with support material on the interface

Note: one supported layer per part. Where a part needs supports, the model keeps its supported surfaces on a single layer, so a printer without an automatic material changer needs only two filament changes for the support material: one before the interface layer and one after. The lid with a bobbin trench and the extension lips each have

one such layer. (The old molded extension pegs added a second, which is one reason the separate pegs are better.) Two supported parts share a plate only when their interfaces are on the same layer, and the plate keeps a corner clear for a purge tower (`purge_tower_size`). The console warns about any part with supports on more than one layer.

7 Assembly

This chapter is an outline, still to be written. It will cover:

- Clearing the membranes and knock-out walls.
- Nuts into the plate and hinge nut traps.
- Joining the halves: dovetails, then the mending plates from below.
- Installing the hinges and setting the machine on its pins.
- Harness rods and carrying.
- Extensions on their pegs, the corner piece's dovetails, and snapping in the lids.

8 Other machines and hinges

This chapter is an outline, still to be written. It will cover:

- What changes between machines, and what to measure.
- Machines without a side compartment, and bases for larger beds.
- Adding a hinge type: the footprint functions and the cutter module.
- Contributing a machine profile.

9 Parameter reference

This chapter is generated from the comments in `scad/vsm_base.scad` by `tools/extract_params.py`. Do not edit it by hand; change the comment in the model and rebuild.

There are 322 parameters; 320 of them appear in the Customizer. The rest are lists the Customizer cannot show, and are marked *edit in the file*.

9.1 Part

Parameter	Default	Description
`part`	`"assembly"`	What to render *Choices: assembly, exploded, all, test_parts, base, base_lid, back_ext, back_lid, left_ext, left_lid, corner_ext, corner_lid, rod_plugs, outlet_cap, hinge_coupon, back_box, left_box, corner_box, ext_pegs, lid_hinge_test, turn_button, passthrough_cover, split_test, bobbin_assembly, bobbin_trays, bobbin_bottom, bobbin_upper, bobbin_test, notion_assembly, notion_set, notion_30, notion_25, notion_20, notion_15, notion_10, notion_custom, spool_tray, spool_assembly, notion_test.*
`explode`	`40`	Spacing between parts in the “exploded” layout
`test_top_h`	`12.7`	Test parts: height of the top slices
`test_floor_extra`	`3`	Test parts: height of the base bottom slice above the cavity floor
`test_coupon_w`	`40`	Test parts: width of the dovetail coupon (X)
`test_slice_w`	`112`	Test parts: width of the center slice (X); covers the plates at 112
`test_stub_w`	`20`	Test parts: full-height wall stubs either side of the joint in the center slice
`test_hinge_w`	`40`	Test parts: size of the hinge block (X), centered on the left hinge
`test_split_gap`	`10`	Gap between the two halves of split test parts
`split_test_h`	`30`	Split test (part split_test): height of the front-wall stubs; 0 = full height *Range 0 to 100, step 1.*
`split_test_w`	`25`	Split test: width either side of the split line *Range 10 to 60, step 1.*
`printer_size`	`[256, 256, 256]`	Printer build volume, for the size check and the print plates
`printer_margin`	`8`	Keep parts this far inside the bed’s edges (room for supports, brims and skirts) *Range 0 to 30, step 1.*
`print_plate`	`0`	Print plates (part all): 0 = every plate, laid out side by side; N = plate N only *Range 0 to 30, step 1.*
`purge_tower_size`	`40`	Purge tower kept clear on plates with supported parts (square, mm); 0 = none *Range 0 to 80, step 5.*
`plate_gap`	`8`	Space between parts on a plate *Range 2 to 30, step 1.*

9.2 Options

Parameter	Default	Description
`compartment_option`	`"storage"`	Small-compartment option (“none” = base fitted to the bed alone) *Choices: none, storage, outlet, knee, jack.*
`hinge_type`	`"lollipop"`	Hinge type *Choices: lollipop.*
`split`	`true`	Split into two dovetailed halves for printing

9.3 Machine

Parameter	Default	Description
`machine`	`"Singer 99K"`	Your machine. A listed machine supplies every setting in the “Custom machine” section; choose Custom to use those settings instead. *Choices: Singer 99K, Custom.*
`bed_cushion_in`	`0.0625`	Bed cushion thickness, compressed, in inches (1/16”)

9.4 Custom machine: IGNORED unless machine = Custom

Parameter	Default	Description
`custom_bed_length_in`	`12.1875`	IGNORED unless machine = Custom. Bed length, inches (99K: 12 3/16”)
`custom_bed_width_in`	`6.5625`	IGNORED unless machine = Custom. Bed width, inches (99K: 6 9/16”)
`custom_bed_plate_thickness_in`	`0.3125`	IGNORED unless machine = Custom. Thickness of the flat bed plate, inches (99K: 5/16”)
`custom_hinge_hole_inset_in`	`1.4375`	IGNORED unless machine = Custom. Hinge hole center from each end of the bed, inches (99K: 1 7/16”)
`custom_hinge_hole_drop_in`	`0.375`	IGNORED unless machine = Custom. Hinge hole center below the bed top surface, inches (99K: 3/8”)
`custom_back_trench`	`[0, 0, 0, 0]`	IGNORED unless machine = Custom. Back bobbin trench, mm: [center from the bed’s left end, width, length back from the bed edge, depth]; all zeros = none
`custom_left_trench`	`[83.34, 57.6, 40, 2.5]`	IGNORED unless machine = Custom. Left bobbin trench, mm: [center from the bed’s front edge, width, length left from the bed edge, depth]; all zeros = none
`custom_base_lid_notch`	`[8, 23]`	IGNORED unless machine = Custom. Notch in the base lid’s left edge clearing the motor boss as the lid opens, mm: [depth, width]; [0, 0] = none

custom_relief_1_edge	"left"	IGNORED unless machine = Custom. Casting relief 1: which edge of the bed the thickened area is on (none = not used) *Choices: none, left, right, back, front.*
custom_relief_1	[83.34, 76.2, 15]	IGNORED unless machine = Custom. Casting relief 1, mm: [center, length, thickness]. Center is along the edge from the bed's left end (back/front edges) or front edge (left/right edges); thickness is measured down from the bed top
custom_relief_2_edge	"right"	IGNORED unless machine = Custom. Casting relief 2: edge *Choices: none, left, right, back, front.*
custom_relief_2	[83.34, 76.2, 15]	IGNORED unless machine = Custom. Casting relief 2, mm: [center, length, thickness]
custom_relief_3_edge	"back"	IGNORED unless machine = Custom. Casting relief 3: edge *Choices: none, left, right, back, front.*
custom_relief_3	[36.51, 50.8, 20]	IGNORED unless machine = Custom. Casting relief 3, mm: [center, length, thickness]
custom_relief_4_edge	"back"	IGNORED unless machine = Custom. Casting relief 4: edge *Choices: none, left, right, back, front.*
custom_relief_4	[273.05, 50.8, 20]	IGNORED unless machine = Custom. Casting relief 4, mm: [center, length, thickness]
custom_relief_5_edge	"none"	IGNORED unless machine = Custom. Casting relief 5: edge *Choices: none, left, right, back, front.*
custom_relief_5	[0, 0, 0]	IGNORED unless machine = Custom. Casting relief 5, mm: [center, length, thickness]
custom_relief_6_edge	"none"	IGNORED unless machine = Custom. Casting relief 6: edge *Choices: none, left, right, back, front.*
custom_relief_6	[0, 0, 0]	IGNORED unless machine = Custom. Casting relief 6, mm: [center, length, thickness]

9.5 Bed seat

Parameter	Default	Description
bed_clearance	0.75	Clearance between bed and seat wall, per side
bed_ledge	8	How far the bed rests on the ledge around the machine cavity *Range 3 to 15, step 0.5.*

rabbit_depth	4	Depth of the bed seat and lid rabbet below the top surface *Range 2 to 8, step 0.1.*

9.6 Body

Parameter	Default	Description
rim_width	9	Top rim: outer face to seat/rabbet edge, front, left and right *Range 5 to 20, step 0.5.*
end_wall	22.225	End wall thickness at the bottom, left and right (7/8” = 22.225)
back_rim_width	0	Top rim: back of the bed seat to outer back face (0 = from the hinge)
hinge_back_skin	5	With automatic back rim: solid behind the hinge’s rear edge *Range 2 to 15, step 0.5.*
bottom_thickness	12.7	Bottom thickness
inner_depth	70	Cavity depth below the top surface
divider_web	4	Solid web at the top between the bed seat and the lid rabbet *Range 2 to 10, step 0.5.*
outer_rounding	5	Outer edge rounding (top and vertical edges) *Range 0 to 10, step 0.5.*

9.7 Small compartment

Parameter	Default	Description
storage_inner_length	70	Inner length of the small compartment (X)
lid_ledge	10	Lid ledge width around the small compartment *Range 5 to 20, step 0.5.*

9.8 Cord pass-through

Parameter	Default	Description
cord_hole_width	46	Cord pass-through width (wide enough for both plugs’ cords side by side)
cord_hole_height	36	Cord pass-through height (without gable)
passthrough_knockouts	false	Close both ends of the pass-through with knock-out walls (tap them out with a punch when you fit the outlet)
passthrough_knockout_t	1.6	Knock-out wall thickness *Range 0.8 to 3, step 0.1.*
passthrough_score_depth	0.6	Score groove around each knock-out wall’s edge, on its tunnel side: depth (deeper = easier to knock out) *Range 0.2 to 1.5, step 0.1.*
passthrough_score_w	0.8	Score groove width *Range 0.4 to 2, step 0.1.*
passthrough_cover	false	Make a removable cover for the pass-through (part passthrough_-cover), pushed into its compartment end
passthrough_cover_depth	6	Cover: depth of the plug into the tunnel *Range 3 to 15, step 0.5.*

9.9 Outlet compartment

Parameter	Default	Description
outlet_size	[76, 40, 22]	Outlet body, mm: [length, width, thickness]. Modern: 76 x 40 x 22; early Bakelite: 83 x 45 x 22
outlet_receptacle_spacing	34	Receptacle spacing, center to center (modern 34, Bakelite 38) *Range 20 to 60, step 0.5.*
outlet_clearance	1.5	Clearance around the outlet inside its curbs, per side
outlet_y_offset	0	Outlet center, front-to-back offset from the middle of the compartment *Range −50 to 50, step 0.5.*
outlet_curb_h	3	Height of the curbs on the floor around the outlet *Range 1 to 8, step 0.5.*
outlet_curb_w	3	Width of the curbs *Range 1 to 8, step 0.5.*
outlet_mount	"through"	How the outlet is held: a heat-set insert in the end wall (needs a soldering iron), or a flat-head screw through the end wall from outside, with a nut on the outlet's face *Choices: through, insert.*
outlet_screw_spec	"#8-32"	Screw through the outlet's center hole (drill out the membrane)
outlet_screw_hole_d	4.6	Through mount: clearance hole for the screw
outlet_screw_head_d	8.4	Through mount: flat-head diameter (#8: 8.4)
outlet_screw_cap	true	Through mount: recess the head under a printed cap (part outlet_cap)
outlet_cap_d	12	Through mount: cap diameter *Range 9 to 20, step 0.5.*
outlet_cap_t	2	Through mount: cap thickness *Range 1 to 4, step 0.1.*
outlet_insert_d	5.6	Hole for the heat-set insert in the right end wall (use the insert maker's recommended diameter)
outlet_insert_depth	12	Depth of the insert hole into the wall *Range 4 to 20, step 0.5.*
outlet_labels	true	Recess the receptacle names into the floor in front of the outlet
outlet_light_side	"back"	Receptacle wired always on (the light); the other is the controlled motor output *Choices: back, front.*
outlet_light_label	"LIGHT"	Name for the always-on receptacle
outlet_motor_label	"MOTOR"	Name for the controlled receptacle
outlet_label_size	7	Letter height *Range 3 to 12, step 0.5.*
outlet_label_depth	0.6	Recess depth *Range 0.2 to 2, step 0.1.*
outlet_lid_hinge	"left"	Edge the outlet-compartment lid hinges on *Choices: left, right, front, back.*
outlet_slot_1_side	"front"	Cord slot 1: lid edge it opens from *Choices: none, front, right, back.*

outlet_slot_2_side	"back"	Cord slot 2: lid edge it opens from *Choices: none, front, right, back.*
outlet_slot_1_pos	0	Slot 1 position along its edge, mm from the lid's left end (front/back edges) or front end (right edge); 0 = automatic
outlet_slot_2_pos	0	Slot 2 position along its edge; 0 = automatic
outlet_slot_w	9	Cord slot width *Range 4 to 20, step 0.5.*
outlet_slot_depth	25	Cord slot depth in from the lid edge
outlet_slot_chamfer	1.5	Chamfer on the slot's top edges, easing a cord pulled over them

9.10 Lollipop hinge

Parameter	Default	Description
hinge_flange_diameter	29	Hinge flange diameter *Range 15 to 50, step 0.1.*
hinge_flange_thickness	3	Hinge flange thickness *Range 1 to 8, step 0.1.*
hinge_thickness	8	Hinge body depth below the top surface *Range 3 to 20, step 0.1.*
hinge_inner_diameter	27	Hinge body diameter *Range 10 to 50, step 0.1.*
hinge_pin_thickness	8	Hinge pin thickness (X) *Range 3 to 15, step 0.1.*
hinge_pin_height	8	Hinge pin height (Z) *Range 3 to 15, step 0.1.*
hinge_pin_extension	25	Hinge pin extension *Range 5 to 60, step 0.5.*
hinge_pin_square_projection	10	Hinge pin square projection *Range 2 to 30, step 0.5.*
hinge_pin_allowed_projection	2	Hinge pin allowed square projection into the bed seat *Range 0 to 6, step 0.1.*
hinge_pin_internal_length	15	Hinge pin internal length *Range 5 to 40, step 0.5.*
hinge_pin_sweep_depth	1	Extra depth below the pin for its corner to sweep through *Range 0 to 4, step 0.1.*
hinge_mounting_hole_diameter	5	Hinge mounting hole diameter *Range 2 to 10, step 0.1.*
hinge_mounting_hole_offset	9.5	Mounting hole center, inward from the flange's back edge

9.11 Hinge coupon

Parameter	Default	Description
hinge_coupon_margin	12	Solid margin around the hinge flange (room for the corner screws)
hinge_coupon_floor	4	Floor left under the deepest hinge cut *Range 2 to 10, step 0.5.*
hinge_coupon_screw_d	4.6	Corner holes for #8 flat-head wood screws: clearance hole
hinge_coupon_head_d	8.4	Corner holes: countersink diameter at the top face

9.12 Hinge nut traps

Parameter	Default	Description
hinge_nut_spec	"#8-32"	Nut for the hinge screws (BOSL2 spec, e.g. “#8-32”, “M4”)
hinge_nut_width	0	Nut width across flats; 0 = from the spec *Range 0 to 20, step 0.05.*
hinge_nut_depth	21	Depth of the nut’s upper face below the top surface *Range 8 to 60, step 0.5.*
hinge_nut_bridge	0.2	Sacrificial bridge over the nut trap (0 = none; drill through after printing)

9.13 Mending plates

Parameter	Default	Description
plate_count	3	Number of plates across the split (0 = none)
plate_length	101.6	Plate length (McMaster 1394A34: 4”)
plate_width	15.875	Plate width (5/8”)
plate_thickness	1.778	Plate thickness (0.070”)
plate_holes	[[-33.866, 2.375], [-11.289, -2.375], [11.289, 2.375], [33.866, -2.375]]	Hole centers [along, across] from the plate center (0.889” pitch, staggered 0.094”) *Edit in the file; not shown in the Customizer.*
plate_hole_bridge	0.3	Sacrificial layer over each screw hole at the pocket ceiling (0 = none); drill through
plate_pocket_clearance	0.5	Clearance around the plate in its pocket, per side
plate_standoff	1.0	How far the plate sits up inside its pocket, so screw heads stay off the table
plate_screw_clearance	4.6	Screw clearance hole diameter (#8)
plate_nut_spec	"#8-32"	Nut for the plate screws (BOSL2 spec)
plate_nut_width	0	Nut width across flats; 0 = from the spec *Range 0 to 20, step 0.05.*
plate_nut_trap_depth	3.5	Nut trap depth into the cavity floor
plate_wall_margin	8	Minimum distance from the cavity walls to the plate edge *Range 2 to 20, step 0.5.*

9.14 Nut fit

Parameter	Default	Description
nut_clearance	0.15	Extra clearance added to nut width (applied to each side)

9.15 Casting reliefs

Parameter	Default	Description
relief_clearance	1	Clearance below the casting in a relief (the reliefs themselves come from the machine profile) *Range 0 to 4, step 0.1.*
relief_margin	2	Extra length added at each end of a relief *Range 0 to 10, step 0.5.*

9.16 Back wall lightening

Parameter	Default	Description
lighten_back_wall	true	Hollow the back wall between the hinges
back_skin	17	Back wall thickness left behind the tunnels *Range 6 to 40, step 0.5.*
back_top_skin	6	Solid material above the tunnel peaks *Range 2 to 20, step 0.5.*
tunnel_width	30	Target tunnel width
tunnel_web	6	Web between tunnels
hinge_tunnel_margin	7.5	Extra solid kept beyond each side of the hinge footprint
split_keepout	15	Solid zone kept on either side of the split, from the cut line

9.17 Split

Parameter	Default	Description
split_part	"both"	Which part to render when split *Choices: both, left, right.*
split_spread	30	Separation between the two halves when both are shown
split_clearance	0.1	Clearance in the joint, per face
key_neck	8	Dovetail key width at the cut line (neck) *Range 4 to 16, step 0.5.*
key_flare_angle	30	Dovetail flare, per side, in degrees; steeper = less pull-apart play
key_depth	8	Dovetail key length, past the cut line into the right half *Range 4 to 16, step 0.5.*
key_ys	[]	Key positions in Y; empty = automatic (floor keys between plate rows, one centered in the back wall) *Edit in the file; not shown in the Customizer.*
serrations	true	Vertical V-serrations across the front wall's joint faces
serration_pitch	3	Serration pitch (ridge to ridge) *Range 2 to 6, step 0.1.*
serration_height	1.5	Serration height (ridge tip beyond the cut line); pitch / 2 gives a 90-degree V
serration_setback	2.5	Solid wall kept between the serrations and each face of the front wall (outer skin and cavity side)
serration_top_gap	3	Serrations stop this far below the bed ledge *Range 1 to 10, step 0.5.*
serration_top_clear	0.5	Extra clearance at the top of each groove, so the ridges never bottom out

9.18 Extensions

Parameter	Default	Description
`back_ext`	`true`	Include the back extension (outfeed support / compartment)
`left_ext`	`true`	Include the left extension
`corner_ext`	`true`	Include the left-rear corner piece (needs both of the above)
`back_ext_length`	`240`	Back extension length (X), from the base's left face
`back_ext_depth`	`100`	Back extension body depth (Y), behind the base
`left_ext_width`	`100`	Left extension body width (X), left of the base
`ext_gap`	`0.3`	Gap between an extension body and the base or another extension
`lip_clearance`	`0.2`	Clearance between the lip and the base top
`ext_top_adjust`	`0`	Trim to the extensions' top plane (lids, lips, trenches) against the real bed, mm; negative lowers it *Range –3 to 3, step 0.1.*
`ext_floor`	`3`	Extension floor thickness *Range 1.5 to 8, step 0.1.*
`ext_frame`	`4`	Frame width around each lid, at the top *Range 2 to 10, step 0.1.*
`ext_ledge`	`4`	Ledge the lid rests on, inside the frame *Range 2 to 10, step 0.1.*
`ext_lid_thickness`	`4`	Lid thickness (= depth of the lid rabbet) *Range 2 to 8, step 0.1.*
`ext_lid_clearance`	`0.3`	Clearance around the lid in its rabbet, per side
`ext_hinge_margin`	`0.5`	Hinged lids: the ledge on the hinge side is widened so the open lid stays this far clear of the bay
`corner_bay_square`	`true`	Keep the corner bay square, with matching extra ledge on its other two sides
`ext_finger_hole`	`22`	Finger hole in each lid
`ext_rounding`	`3`	Rounding of the extensions' exposed edges *Range 0 to 8, step 0.5.*

9.19 Turn button

Parameter	Default	Description
`turn_button`	`false`	Turn button on the front rim that holds the bed's front edge down (off by default)
`turn_button_x`	`0`	Position along the front rim (X); 0 = automatic, between the split and the compartment
`turn_button_overlap`	`4`	How far the button reaches over the bed's front edge *Range 1 to 10, step 0.5.*
`turn_button_w`	`9`	Button width *Range 6 to 16, step 0.5.*
`turn_button_t`	`4`	Button thickness *Range 2 to 8, step 0.5.*
`turn_button_gap`	`0.3`	Gap between the button and the bed top
`turn_button_ease`	`10`	Easing: how far the pad's smooth slopes run out along the rim, each side

`turn_button_swell`	`3`	How far the pad swells out of the front face, enclosing the nut *Range 0 to 8, step 0.5.*
`turn_button_swell_taper`	`10`	Below the nut, the swelling blends back into the front face over this height
`turn_button_nut_spec`	`"#8-32"`	Nut for the button screw (BOSL2 spec)

9.20 Freestanding boxes

Parameter	Default	Description
`make_back_box`	`false`	Also make a freestanding version of the back extension (part back_box): same bay and lid, no lip, pegs or dovetails
`make_left_box`	`false`	Also make a freestanding left box (part left_box)
`make_corner_box`	`false`	Also make a freestanding corner box (part corner_box)

9.21 Extension pegs

Parameter	Default	Description
`ext_peg_holes`	`true`	Holes in the base rim for the extension pegs
`ext_peg_style`	`"insert"`	Pegs: insert (separate printed pegs, part ext_pegs, printed lying down), pin (cut from rod_d rod), or molded into the lip (as before v20) *Choices: insert, pin, molded.*
`peg_d`	`4`	Peg diameter (printed pegs) *Range 3 to 8, step 0.1.*
`peg_len`	`3`	Peg length below the lip *Range 1.5 to 6, step 0.1.*
`peg_hole_d`	`4.4`	Hole diameter in the base (for printed pegs; pins use rod_d + 0.2)
`peg_lip_clearance`	`0.2`	Peg hole in the lip, over the peg diameter (the peg's crush ribs make the press fit)
`pin_lip_interference`	`0.1`	Metal pins: press fit in the lip, hole under the rod diameter
`peg_hole_depth`	`3.5`	Hole depth in the base
`back_peg_xs`	`[16, 230]`	Back-rim peg positions (X); keep clear of the hinges and the split
`left_peg_ys`	`[35, 150]`	Left-rim peg positions (Y)
`right_peg_ys`	`[35, 150]`	Right-rim peg positions (Y), for future right-side extensions; [] for none

9.22 Corner dovetails

Parameter	Default	Description
`ext_dt_neck`	`6`	Dovetail width at the mating face *Range 3 to 12, step 0.5.*
`ext_dt_head`	`9`	Dovetail width at its head *Range 4 to 16, step 0.5.*
`ext_dt_depth`	`4`	Dovetail depth into the mating wall *Range 2 to 10, step 0.5.*
`ext_dt_clearance`	`0.2`	Dovetail clearance per face

9.23 Handholds

Parameter	Default	Description
handholds	true	Handholds under each end
handhold_bottom_len	76.2	Handhold length along the end, at the bottom face (3”)
handhold_top_len	50.8	Handhold length at its top (2”)
handhold_height	12.7	Handhold height above the bottom face (1/2”)
handhold_depth	15.875	Handhold depth into the end wall (5/8”)
handhold_lip_flat	4	Flat underside of the lip at the face, before the ceiling climbs *Range 0 to 10, step 0.5.*
handhold_lip_round	2	Radius rounding the lip’s outer edge *Range 0 to 6, step 0.5.*
handhold_knockout	true	Knock-out wall under the lip, removed after printing
knockout_t	1.2	Knock-out wall thickness
knockout_gap	0.2	Gap between the knock-out wall and the lip
knockout_tab	true	Breakaway tab tying one end of the knock-out wall to the pocket end
knockout_tab_h	0.8	Tab height (the tab sits just under the lip gap)
knockout_brim	true	Brim around the foot of each knock-out wall, so it stands until the tab joins it
knockout_brim_w	5	Brim width each side of the wall
knockout_brim_h	0.2	Brim height (about one first layer)

9.24 Harness rods

Parameter	Default	Description
harness_rods	true	Rod notches and rods either side of each handhold
rod_d	4.7625	Rod diameter (3/16”)
rod_hole_d	4.95	Rod hole diameter (printed as a teardrop, point up)
rod_x	7	Rod center in from the end face *Range 3 to 15, step 0.1.*
rod_z	7	Rod center above the bottom face *Range 3 to 15, step 0.1.*
notch_len	25	Notch length along the rod
notch_clear	4	Clear space around the rod in the notch, behind and above it *Range 1 to 10, step 0.5.*
notch_inset	25	Distance from each outer face (front, back) to its notch
rod_stop	5	Material left between a rod’s blind end and the handhold
rod_plug_len	5	Printed plug closing the open end of each rod hole (0 = no plugs; rods run to the face) *Range 0 to 10, step 0.5.*
rod_plug_clearance	0.1	Plug clearance in the hole, per side (the crush ribs make the press fit)

`rod_plug_rib`	`0.15`	Crush rib interference, beyond the hole wall

9.25 Left lip

Parameter	Default	Description
`lip_edge_round`	`2`	Radius of the rounded top edge along the lip front (also the free edges of an extended back lip) *Range 0 to 5, step 0.5.*
`lip_fillet`	`5`	Radius of the inside corner where the lip front meets the body (also an extended back lip's corners) *Range 0 to 15, step 0.5.*

9.26 Back lip

Parameter	Default	Description
`back_lip_extend`	`true`	Run the back lip on past a short back extension, over the base's back rim, to reach the last back peg hole
`back_lip_peg_margin`	`10`	How far the lip runs past the last peg's center *Range 6 to 30, step 0.5.*

9.27 Lid hinges

Parameter	Default	Description
`base_lid_hinge`	`"right"`	Lid edge each lid hinges on *Choices: left, right, front, back.*
`back_lid_hinge`	`"back"`	*Choices: left, right, front, back.*
`left_lid_hinge`	`"left"`	*Choices: left, right, front, back.*
`corner_lid_hinge`	`"back"`	*Choices: left, right, front, back.*
`base_lid_type`	`"hinged"`	Base compartment lid: hinged (snap-in dimple hinge), lift-out (plain lid resting on the ledge), or none (rabbet left for a lid later) *Choices: hinged, lift-out, none.*
`back_lid_type`	`"hinged"`	Back extension lid type *Choices: hinged, lift-out, none.*
`left_lid_type`	`"hinged"`	Left extension lid type *Choices: hinged, lift-out, none.*
`corner_lid_type`	`"hinged"`	Corner piece lid type *Choices: hinged, lift-out, none.*
`base_lid_style`	`"solid"`	Base compartment lid style: solid, or perforated for ventilation *Choices: solid, perforated.*
`back_lid_style`	`"solid"`	Back extension lid style *Choices: solid, perforated.*
`left_lid_style`	`"solid"`	Left extension lid style *Choices: solid, perforated.*
`corner_lid_style`	`"solid"`	Corner piece lid style *Choices: solid, perforated.*
`base_lid_grip`	`"finger hole"`	Base compartment lid grip: finger hole, or a thumb tab reaching over a notch in the wall *Choices: finger hole, tab.*
`base_lid_tab_edge`	`"front"`	Base lid tab edge (front only is practical, on a lift-out lid) *Choices: front, back, left, right.*
`back_lid_grip`	`"finger hole"`	Back extension lid grip *Choices: finger hole, tab.*

back_lid_tab_edge	"right"	Back lid tab edge (back or right) *Choices: front, back, left, right.*
left_lid_grip	"finger hole"	Left extension lid grip *Choices: finger hole, tab.*
left_lid_tab_edge	"front"	Left lid tab edge (front or left) *Choices: front, back, left, right.*
corner_lid_grip	"finger hole"	Corner piece lid grip *Choices: finger hole, tab.*
corner_lid_tab_edge	"left"	Corner lid tab edge (left or back) *Choices: front, back, left, right.*
grip_tab_w	25	Thumb tab width
grip_tab_over	1.5	Thumb tab overhang past the outer face
grip_tab_r	2	Thumb tab corner radius *Range 0 to 5, step 0.5.*
grip_scoop_d	2.5	Thumb scoop below the tab: depth into the outer face
grip_scoop_h	6	Thumb scoop height below the lid *Range 2 to 12, step 0.5.*
base_lid_notch_r	4	Base lid motor-boss notch (size from the machine profile): inner corner radius *Range 0 to 10, step 0.5.*
base_lid_notch_flare	10	Notch: radius blending it into the lid edge
base_lid_notch_offset	0	Notch: front-to-back offset from the middle of the lid *Range –50 to 50, step 0.5.*
lid_perf_d	5	Perforation hole diameter *Range 2 to 12, step 0.5.*
lid_perf_pitch	8	Perforation pitch (hex grid) *Range 4 to 20, step 0.5.*
lid_perf_margin	6	Solid margin kept around the lid's edges, finger hole and hinge *Range 2 to 20, step 0.5.*
lid_bump_d	3	Hinge bump diameter at its base *Range 1.5 to 6, step 0.1.*
lid_bump_engage	1.0	How far each bump engages the wall
lid_tab_len	15	Flex tab carrying each bump: length along the end
lid_tab_w	4	Flex tab width *Range 1.5 to 8, step 0.1.*
lid_slot_w	1	Slot that frees the flex tab *Range 0.5 to 3, step 0.1.*
lid_bump_angle	35	Bump flank angle from its axis: smaller is steeper and holds better (45 = the v19 cone) *Range 15 to 60, step 1.*
lid_hinge_end_clearance	0.15	Clearance at the lid's two hinge ends (less end float keeps both bumps engaged)
lid_recess_clearance	0.075	Clearance between bump and recess, per side
lid_open_stop	true	Cut a chamfer behind the hinge so an open lid leans back and stays up

9.28 Inserts

Parameter	Default	Description
ins_spacing	10	Spacing between parts in the insert print layouts *Range 0 to 50, step 1.*

`bobbin_d`	`20.6375`	Bobbin diameter (class 15 and 66: 13/16”), for the bobbin trays and the spool tray
`bobbin_t`	`11.1125`	Bobbin thickness (class 15 and 66: 7/16”)

9.29 Bobbin trays (corner bay)

Parameter	Default	Description
`bb_trays`	`0`	Number of trays, bottom included; 0 = as many as fit under the lid *Range 0 to 10, step 1.*
`bb_post_d`	`5`	Post diameter (class 15 and 66 bores are a little over 6 mm) *Range 3 to 10, step 0.1.*
`bb_grid`	`[3, 3]`	Post grid [columns (X), rows (Y)]; cells that would crowd the center column are left empty
`bb_floor`	`3`	Tray floor thickness *Range 1.2 to 6, step 0.1.*
`bb_bobbin_clear`	`1`	Clearance between a bobbin’s top and the tray above *Range 0 to 4, step 0.1.*
`bb_wall_clear`	`0.75`	Clearance between the trays and the bay walls, per side *Range 0.2 to 3, step 0.05.*
`bb_corner_r`	`2`	Tray corner radius *Range 0 to 10, step 0.5.*
`bb_top_clear`	`1`	Clearance between the top of the knob and the lid *Range 0 to 5, step 0.1.*
`bb_lighten`	`true`	Lightening holes where four cells meet
`bb_lighten_d`	`12`	Lightening hole diameter *Range 4 to 20, step 0.5.*
`bb_pin_d`	`3`	Pin on each post tip: diameter *Range 1.5 to 6, step 0.1.*
`bb_pin_h`	`2`	Pin height above the post shoulder *Range 0.5 to 5, step 0.1.*
`bb_socket_clear`	`0.2`	Socket clearance around the pin, per side *Range 0 to 1, step 0.05.*
`bb_socket_extra`	`0.4`	Socket depth beyond the pin *Range 0 to 2, step 0.1.*
`bb_socket_chamfer`	`0.6`	Chamfer at the socket mouth (also absorbs elephant’s foot) *Range 0 to 2, step 0.1.*
`bb_column_d`	`12`	Handle column diameter *Range 6 to 20, step 0.5.*
`bb_column_flare`	`2`	Flare radius at the column’s foot *Range 0 to 5, step 0.5.*
`bb_knob_d`	`18`	Knob diameter *Range 8 to 30, step 0.5.*
`bb_knob_h`	`8`	Knob height *Range 3 to 20, step 0.5.*
`bb_hole_clear`	`0.5`	Clearance between the knob and the upper trays’ center holes, per side *Range 0.1 to 2, step 0.05.*

9.30 Notion trays (back or left bay)

Parameter	Default	Description
`ot_bay`	`"back"`	Bay the notion and spool trays go in *Choices: back, left.*
`ot_custom_h`	`12`	Custom tray height *Range 5 to 80, step 0.5.*

`ot_show`	`[25, 20, 15, 10]`	Tray heights shown stacked in part notion_assembly
`ot_wall_clear`	`1`	Clearance between the trays and the bay walls, per side *Range 0.2 to 3, step 0.05.*
`ot_wall`	`1.2`	Wall thickness *Range 0.8 to 4, step 0.1.*
`ot_floor`	`1.6`	Floor thickness *Range 0.8 to 4, step 0.1.*
`ot_corner_r`	`2`	Outer corner radius *Range 0 to 10, step 0.5.*
`ot_boss`	`9`	Corner boss size (square, from the outer corner) *Range 5 to 20, step 0.5.*
`ot_pin_d`	`3`	Pin diameter *Range 1.5 to 6, step 0.1.*
`ot_pin_h`	`2`	Pin height above the wall tops *Range 0.5 to 5, step 0.1.*
`ot_socket_clear`	`0.2`	Socket clearance around the pin, per side *Range 0 to 1, step 0.05.*
`ot_socket_extra`	`0.4`	Socket depth beyond the pin *Range 0 to 2, step 0.1.*
`ot_socket_chamfer`	`0.5`	Chamfer at the socket mouth (also absorbs elephant's foot) *Range 0 to 2, step 0.1.*
`ot_grip_w`	`40`	Finger-hold lip length along each end wall *Range 10 to 120, step 1.*
`ot_grip_reach`	`5`	How far the lip reaches in from the wall *Range 0 to 15, step 0.5.*
`ot_grip_t`	`1.5`	Thickness of the lip at its top, above the 45-degree underside *Range 0.8 to 4, step 0.1.*
`ot_top_clear`	`1`	Clearance between the top tray's pins and the lid *Range 0 to 5, step 0.1.*

9.31 Notion tray compartments

Parameter	Default	Description
`ot_compartments`	`false`	Divide the plain notion trays (not the spool tray) into compartments
`ot_cell`	`25.4`	Target compartment size; the grid divides the tray evenly to come closest to it *Range 10 to 100, step 0.1.*
`ot_cols`	`0`	Compartments along the tray's length (0 = from ot_cell) *Range 0 to 20, step 1.*
`ot_rows`	`0`	Compartments across the tray's width (0 = from ot_cell) *Range 0 to 20, step 1.*
`ot_divider_frac`	`0.8`	Divider height, as a fraction of the tray height *Range 0.2 to 1, step 0.05.*
`ot_divider_t`	`1.2`	Divider thickness *Range 0.8 to 4, step 0.1.*

9.32 Spool tray

Parameter	Default	Description
`ot_spool_d`	`38`	Spool diameter *Range 20 to 80, step 0.5.*
`ot_spool_h`	`44`	Spool height *Range 20 to 100, step 0.5.*
`ot_spool_post_d`	`5`	Post diameter at the top, through the bobbin *Range 3 to 15, step 0.1.*

`ot_spool_post_base_d`	`7`	Post diameter at the floor; the post tapers to the top diameter at the spool top *Range 3 to 20, step 0.1.*
`ot_spool_bore`	`7.5`	Smallest spool bore you expect (for the console check) *Range 3 to 20, step 0.1.*
`ot_spool_gap`	`1.5`	Space between neighboring spools *Range 0 to 10, step 0.1.*
`ot_spool_wall_gap`	`0.5`	Space between the spools and the tray walls and corner bosses *Range 0 to 5, step 0.1.*
`ot_spool_top_clear`	`1`	Clearance between the bobbin tops and the tray above *Range 0 to 5, step 0.1.*
`ot_spool_open_sides`	`true`	Cut the long side walls down to a curb between the corner bosses, so the spools can be grasped from the side
`ot_spool_curb`	`10`	Curb height left along the long sides (it stiffens the floor when the tray is lifted by its ends) *Range 3 to 40, step 0.5.*
`ot_spool_cut_r`	`5`	Radius at the bottom corners of the cutaway *Range 0 to 20, step 0.5.*